

ECON 636 600: MACROECONOMIC THEORY I

Assignment 2 **Solution**

1. (30 points) Suppose there is a consumer with utility

$$\sum_{t=0}^{\infty} \beta^t u(c_t).$$

The consumer has a sequence of income $\{y_t\}_{t=0}^{\infty}$, and a sequence of budget constraints

$$c_t + b_{t+1} \leq Rb_t + y_t,$$

where b_t is the holding of bonds (positive holding is saving, while negative holding is borrowing) and $\beta R = 1$. Initially the consumer has neither savings nor borrowing (i.e., $b_0 = 0$). The consumer is subject to a non-Ponzi game condition:

$$\lim_{T \rightarrow \infty} R^{-T} b_{T+1} \geq 0.$$

Find the consumer's optimal consumption and savings path.

The consolidated budget constraint is

$$\sum_{t=0}^T R^{-t} c_t + R^{-T} b_{T+1} = \sum_{t=0}^T R^{-t} y_t,$$

which, after taking limit $T \rightarrow \infty$, is

$$\sum_{t=0}^{\infty} R^{-t} c_t + \lim_T R^{-T} b_{T+1} = \sum_{t=0}^{\infty} R^{-t} y_t.$$

The non-ponzi-game condition requires $\lim_T R^{-T} b_{T+1} \geq 0$, hence the budget set of the consumer is

$$\sum_{t=0}^{\infty} R^{-t} c_t \leq \sum_{t=0}^{\infty} R^{-t} y_t.$$

The Lagrangian is

$$L = -\sum_{t=0}^{\infty} \beta^t u(c_t) + \lambda \left(\sum_{t=0}^{\infty} R^{-t} c_t - \sum_{t=0}^{\infty} R^{-t} y_t \right),$$

with first order conditions

$$\frac{\partial L}{\partial c_t} = -\beta^t u'(c_t) + R^{-t} \lambda = 0.$$

Hence

$$u'(c_t) = u'(c_s), \forall t \neq s.$$

To find out the constant level of consumption (denoted as $\bar{c}$), use the budget constraint. The budget constraint is binding because $\lambda > 0$.

$$\sum_{t=0}^{\infty} R^{-t} \bar{c} = \sum_{t=0}^{\infty} R^{-t} y_t.$$

hence $\bar{c} = (1 - R^{-1}) \sum_{t=0}^{\infty} R^{-t} y_t$. Using this formula for consumption, we can find the asset holding b_t to be

$$\begin{aligned} \sum_{s=0}^{\infty} R^{-s} y_s &= \sum_{s=t}^{\infty} R^{-(s-t)} c_s = R b_t + \sum_{s=t}^{\infty} R^{-(s-t)} y_s, \\ b_t &= \frac{\sum_{s=0}^{\infty} R^{-s} y_s - \sum_{s=t}^{\infty} R^{-(s-t)} y_s}{R}. \end{aligned}$$

We can verify that

$$b_t = \sum_{s=0}^t R^{t-s} (y_t - \bar{c}).$$

2. (40 points) Consider a social planner in charge of the production of two firms. Firm 1 has a production technology $f(k_1, n_1) = k_1^{1/3} n_1^{2/3} + k_1$, while firm 2 has $g(k_2, n_2) = k_2^{2/3} n_2^{1/3} + n_2$. Suppose the total supply of capital is 5 ($k_1 + k_2 = 5$), and the total supply of labor is 3 ($n_1 + n_2 = 3$).

- (a) Formulate the social planner's output maximization problem. Derive the Kuhn-Tucker conditions for the problem. (20 points)

The social planner's problem is

$$\begin{aligned} \max \quad & k_1^{1/3} n_1^{2/3} + k_1 + k_2^{2/3} n_2^{1/3} + n_2 \\ \text{s.t.} \quad & k_1 + k_2 = 5 \\ & n_1 + n_2 = 3. \end{aligned}$$

Lagrangian

$$L = -[k_1^{1/3} n_1^{2/3} + k_1 + k_2^{2/3} n_2^{1/3} + n_2] + \lambda_1(k_1 + k_2 - 5) + \lambda_2(n_1 + n_2 - 3),$$

with KT conditions

$$\begin{aligned} \frac{\partial L}{\partial k_1} &= -\left[\frac{1}{3}k_1^{-\frac{2}{3}}n_1^{\frac{2}{3}} + 1.0\right] + \lambda_1 = 0 \\ \frac{\partial L}{\partial k_2} &= -\left[\frac{2}{3}k_2^{-\frac{1}{3}}n_2^{\frac{1}{3}}\right] + \lambda_1 = 0, \\ \frac{\partial L}{\partial n_1} &= -\left[\frac{2}{3}k_1^{\frac{1}{3}}n_1^{-\frac{1}{3}}\right] + \lambda_2 = 0 \\ \frac{\partial L}{\partial n_2} &= -\left[1.0 + \frac{1}{3}k_2^{\frac{2}{3}}n_2^{-\frac{2}{3}}\right] + \lambda_2 = 0. \end{aligned}$$

- (b) Show that the KT conditions coincide with the equilibrium conditions in Question 2, Assignment 1. This means that the first and the second welfare theorems hold in this environment as well. (20 points)

If we take $\lambda_1 = R$ and $\lambda_2 = W$, then the K-T condition coincides with the equilibrium conditions.

$$\begin{aligned} \frac{1}{3}k_1^{-\frac{2}{3}}n_1^{\frac{2}{3}} + 1.0 &= R = \frac{2}{3}k_2^{-\frac{1}{3}}n_2^{\frac{1}{3}}, \\ \frac{2}{3}k_1^{\frac{1}{3}}n_1^{-\frac{1}{3}} &= W = 1.0 + \frac{1}{3}k_2^{\frac{2}{3}}n_2^{-\frac{2}{3}}, \end{aligned}$$

3. (30 points) Consider an unemployment worker with a discount factor β who is searching for a job. Employment provides a wage income $w > 0$, while unemployment provides 0 income. Job finding probability is $p(a_t)$, where $p(\cdot) : [0, \infty) \rightarrow [0, 1]$ is an increasing function, and a_t is the search effort of the worker. For simplicity, assume that an employed worker never loses the job, and his discounted utility is $\frac{u(w)}{1-\beta}$. An unemployed worker's period utility is $u(0) - a_t$, i.e., utility from consuming 0 minus the disutility from searching. Carefully rewrite the decision problem as a dynamic programming problem. Please be precise about the state variable and the control variable.

State space= $\{E, U\}$, Employment and Unemployment. The return function is

$$r(s, a) = \begin{cases} u(w), & s = E, \\ u(0) - a, & s = U \end{cases}$$

and the functional equations are

$$\begin{aligned} V(E) &= r(E, a) + \beta V(E), \\ V(U) &= \max_{a \geq 0} \{r(U, a) + \beta[p(a)V(E) + (1 - p(a))V(U)]\}. \end{aligned}$$